

SHERBAKUL', Russian Federation

WMO: 287910

Lat: 54.633N

Lon: 72.433E

Elev: 115

StdP: 99.95

Time Zone: 6.00 (E06)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -34.4	(c) -31.4	(d) -37.2	(e) 0.1	(f) -34.2	(g) -34.2	(h) 0.2	(i) -31.2	(j) 10.4	(k) -8.7	(l) 9.4	(m) -6.8	(n) 1.6	(o) 220	(p) 0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	30.6	18.5	28.7	18.1	27.0	17.7	20.5	27.0	19.6	25.9	18.8	24.7	4.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.4	13.4	23.1	17.5	12.7	22.3	16.5	11.9	21.5	59.8	26.7	56.7	25.9	53.9	24.7	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.0	8.7	7.8	-36.3	33.9	4.2	1.7	-39.3	35.2	-41.7	36.2	-44.0	37.1	-47.0	38.4
			-36.5	22.0	3.9	0.8	-39.3	22.6	-41.6	23.1	-43.7	23.5	-46.5	24.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.9	-18.4	-16.0	-7.7	4.1	12.6	18.2	18.9	16.7	10.8	3.4	-6.6	-14.6	(6)	(7)
	DBStd	14.61	7.97	7.20	6.22	5.92	5.00	4.28	3.65	3.85	4.45	4.78	7.38	7.97	(7)	(8)
	HDD10.0	3897	879	729	549	191	32	2	0	2	42	209	498	763	(8)	(9)
	HDD18.3	6155	1138	962	808	427	186	54	39	77	230	464	748	1021	(9)	(10)
	CDD10.0	931	0	0	0	14	112	249	276	211	66	3	0	0	(10)	(11)
	CDD18.3	148	0	0	0	0	8	52	57	27	4	0	0	0	(11)	(12)
	CDH23.3	1570	0	0	0	6	152	553	512	282	66	0	0	0	(12)	(13)
	CDH26.7	448	0	0	0	1	32	181	144	76	15	0	0	0	(13)	(14)
(14) Wind	WSAvg	3.5	3.4	3.5	4.1	4.0	3.9	3.3	3.0	3.0	3.2	3.5	3.8	3.8	(14)	(15)
(15) Precipitation	PrecAvg	361	16	14	16	22	31	43	63	55	22	29	27	20	(15)	(16)
	PrecMax	510	39	38	42	52	122	87	124	123	52	71	72	41	(16)	(17)
	PrecMin	217	0	1	2	4	2	13	12	17	7	2	6	4	(17)	(18)
	PrecStd	73	9	7	10	12	22	16	33	26	13	18	15	10	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.2	1.0	6.7	23.4	30.6	32.8	33.1	31.7	28.6	18.7	7.1	1.4	(19)	(20)
		MCWB	-0.3	0.0	4.0	12.8	16.2	18.2	19.7	18.8	15.8	10.9	5.5	0.2	(20)	(21)
	2%	DB	-2.0	-0.8	3.0	19.5	27.4	30.7	30.0	28.6	25.1	16.2	5.2	0.2	(21)	(22)
		MCWB	-2.6	-1.5	1.4	10.6	14.7	18.4	19.1	18.3	14.2	9.7	4.1	-0.6	(22)	(23)
	5%	DB	-3.9	-2.6	1.9	17.0	25.0	28.8	28.0	26.4	21.7	13.4	3.5	-1.6	(23)	(24)
		MCWB	-4.4	-3.2	0.7	9.8	14.1	18.2	18.9	17.9	13.7	8.4	2.3	-2.3	(24)	(25)
	10%	DB	-6.1	-4.9	0.8	13.9	22.4	26.6	26.2	24.4	18.9	10.9	1.7	-4.0	(25)	(26)
		MCWB	-6.6	-5.5	-0.2	8.5	13.0	17.5	18.5	17.0	12.2	7.3	0.7	-4.8	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.3	0.1	4.1	13.8	17.4	21.6	21.6	21.0	17.6	12.4	6.0	0.6	(27)	(28)
		MCDB	0.3	0.9	6.4	21.5	26.8	28.8	27.4	26.9	25.9	17.2	6.7	1.0	(28)	(29)
	2%	WB	-2.6	-1.5	1.7	11.7	15.9	20.2	20.7	19.7	15.6	10.5	4.2	-0.5	(29)	(30)
		MCDB	-2.1	-0.9	2.7	17.8	25.2	27.5	26.9	25.4	22.5	14.6	5.2	0.2	(30)	(31)
	5%	WB	-4.4	-3.2	0.9	10.1	14.7	19.1	19.9	18.7	14.1	9.0	2.3	-2.2	(31)	(32)
		MCDB	-3.9	-2.6	1.8	16.2	22.7	25.8	26.1	24.6	20.3	12.7	3.3	-1.5	(32)	(33)
	10%	WB	-6.6	-5.5	-0.2	8.7	13.7	18.0	19.1	17.8	12.7	7.5	0.8	-4.7	(33)	(34)
		MCDB	-6.1	-4.9	0.8	13.9	21.1	25.3	24.8	23.2	18.0	10.7	1.5	-4.1	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	7.7	8.4	8.5	9.9	12.2	11.3	10.1	10.5	10.6	8.1	6.0	7.5	(35)	(36)
		MCDBR	7.7	7.0	6.6	15.7	16.5	14.9	13.4	14.7	15.7	13.6	6.3	7.4	(36)	(37)
		MCWBR	7.6	6.8	5.8	8.2	6.8	5.7	5.1	6.3	7.1	7.9	5.5	7.3	(37)	(38)
	5% WB	MCDBR	7.6	7.0	6.7	14.3	14.2	12.6	11.2	11.9	13.6	11.7	5.9	7.6	(38)	(39)
		MCWBR	7.5	6.9	6.0	8.1	6.9	5.6	4.9	5.6	6.7	7.4	5.4	7.4	(39)	(40)
(40) Clear-Sky Solar Irradiance	taub		0.286	0.317	0.375	0.408	0.404	0.409	0.435	0.432	0.371	0.353	0.324	0.290	(40)	(41)
	taud		2.048	2.018	1.943	2.129	2.223	2.293	2.207	2.183	2.313	2.242	2.108	2.014	(41)	(42)
	Ebn at Noon		617	733	769	810	839	839	805	780	786	697	562	509	(42)	(43)
	Edh at Noon		70	103	141	135	130	123	132	127	98	83	66	58	(43)	(44)
(44) All-Sky Solar Radiation	RadAvg		1.02	2.11	3.67	4.66	5.70	6.05	5.55	4.52	3.24	1.69	0.81	0.66	(44)	(45)
	RadStd		0.07	0.09	0.31	0.38	0.49	0.33	0.41	0.34	0.24	0.22	0.11	0.08	(45)	(46)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page